

Table 1. The Solar System forms from a spinning cloud of gas and dust inside a dense stellar cluster containing around 1000 neighboring stars. Timeline and overview of the most important events.

10 ⁹ a	Inner Solar System					Outer Solar System					Hills cloud	Oort cloud
	<i>Mercury</i> 10 ⁹ a event	<i>Venus(?)</i> 10 ⁹ a event	<i>Earth</i> 10 ⁹ a event	<i>Mars</i> 10 ⁹ a event	<i>Asteroid belt</i> 10 ⁹ a event	<i>Jupiter</i> 10 ⁹ a event	<i>Saturn</i> 10 ⁹ a event	<i>Uranus</i> 10 ⁹ a event	<i>Neptune</i> 10 ⁹ a event	<i>Kuiper belt</i> 10 ⁹ a event		
4,7												
4,6												
4,5	4,5 Forms ¹	4,56 Forms	4,54 Forms	4,6 Forms	4,56 Protoplanetary Disk ²⁰	4,6 Forms ²³ , Grand Tack ¹⁴	4,59 Forms	4,59 Forms ¹⁸	4,59 Forms	4,6 Solar nebula	4,6 Begins forming ¹⁴	
4,4	4,5-3,8 <i>Pre-Tolstojan & Tolstojan Eras</i> : Solid crust	4,4 Protoplanet collides ⁵ 4,39 Crashes back ⁶ 4,3-0,7 Habitable ⁷	4,5 Impact, Moon ⁹ 4,4-4 <i>Hadean Eon</i>	4,5-4,1 <i>Pre-Noachian Era</i> : Crust solidified rapidly ¹⁶	4,5-4,1 <i>Primordial Sorting & The Grand Tack</i> ²¹	4,5-4,1 Four largest Galilean moons ²⁵		4,52 Collision, axial tilt retrograde rotation ²⁹	4,5 Migrate outward	4,5 Fail to coalesce ³² 4,4 Scattered disc ³³	4,5 Stable inner disk ³⁵ 4,5- Steady-state buffer ³⁶	4,5 <i>Galactic ejection</i> ³⁷
4,3												
4,2												
4,1												
4,0												
3,9	Heavily battered			4,1-3,7 <i>Noachian Period</i> : Rivers and lakes ¹⁷	4,1-3,8 <i>Mass Depletion</i> ²²	4,1-3,8 Giants shift <i>Late Heavy Bombardment</i> ²⁶	4,1-3,8 Giants shift <i>Late Heavy Bombardment</i>	4,1-3,8 Giants shift ³⁰ <i>Late Heavy Bombardment</i>		4 Objects stabilize		
3,8	3,8 <i>Calorian Era</i> ²		4-2,5 <i>Archean Eon</i> : 3,85 <i>PRE-GOE</i> ¹⁰									
3,7				3,7-3,0 <i>Hesperian Period</i> : Core cooling, loses its global magnetic field ¹⁸								
3,6					3,5- <i>Slow Attrition</i>							
3,5	3,5 Core cools ³ 3,5- <i>Mansurian & Kuiperian Eras</i> ⁴											
3,4												
3,3												
3,2												
3,1												
3,0			3 Cyanobacteria ¹¹									
2,9												
2,8												
2,7												
2,6												
2,5												
2,4												
2,3												
2,2												
2,1												
2,0												
1,9												
1,8												
1,7												
1,6												
1,5			1,6 Plant-like organisms									
1,4												
1,3												
1,2												
1,1												
1,0												
0,9												
0,8												
0,7												
0,6		0,7- Greenhouse effect ⁸	0,8-0,7 First Animals <i>Snowball Earth</i> ¹² 0,63 Bilateral symmetry							0,6 Inner ring system ³¹		
0,5			0,54- <i>Phanerozoic Eon</i> : Cambrian Explosion ¹³									
0,4			0,252 Pangea forms									
0,3			0,25 <i>Great Dying</i> ¹⁴				0,2-0,1 Ring system forms ²⁷					
0,2												
0,1												
0,0			0,066 <i>Cenozoic Era</i> ¹⁵									

Note:

¹) Planetesimals collide and coalesce to form Mercury, a vast magma ocean covers its surface.

²) A massive asteroid impact creates the 'Caloris Basin', one of the Solar System's largest impact basins, magma floods the plains.

³) Volcanic 'shutoff': Mercury's core cools and contracts, causing the crust to wrinkle with massive cliffs and sealing off volcanic vents.

⁴) Slow, intermittent space weathering and occasional meteoroid impacts.

⁵) A massive protoplanet collides with Venus, temporarily granting it a moon.

⁶) The moon orbit decays and crashes back into Venus, this catastrophic impact (likely) reverses its rotation, creating its slow, retrograde spin.

⁷) Liquid water oceans and early plate tectonics potentially create a habitable, Earth-like environment for up to 3 billion years.

⁸) Global volcanic activity releases massive amounts of carbon dioxide into the atmosphere. Without active plate tectonics or water to trap carbon, a runaway 'greenhouse effect' takes hold, oceans boil away completely.

⁹) A giant protoplanet collision 'tilted' Earth's axis, created 24-hour days and formed the Moon from the broken debris.

¹⁰) No O₂ in the atmosphere.

¹¹) First to perform photosynthesis, leading to the 'Great Oxygenation Event', completely transforming Earth's atmosphere.

¹²) Drastic climate drops encased the entire planet in ice multiple times, forcing surviving organisms to adapt to extreme conditions.

¹³) A massive burst of evolution occurs, establishing nearly all modern animal phyla, including mollusks, arthropods and chordates.

¹⁴) Triassic Era, after the 'Great Dying' extinction (the Permian-Triassic extinction event), the first dinosaurs and early mammals evolve.

¹⁵) The modern Era, a 'mass extinction' (the Cretaceous-Paleogene extinction event K-Pg) wipes out non-avian dinosaurs, allowing mammals to rapidly diversify and become dominant.

¹⁶) The planet's crust solidified rapidly around 4.54 billion years ago. The planetary 'dichotomy' (the northern lowlands and southern highlands) formed via massive asteroid impacts during this phase.

¹⁷) The planet possessed a thick atmosphere, a protective magnetic field and extensive surface rivers and lakes.

¹⁸) Global core cooling caused the planet to lose its global magnetic field, subsurface water releases carved immense outflow channels across the surface.

¹⁹) Mars dried out completely into an arid desert, dominated by iron oxide dust storms and glacier-like ice movements. Mars transitioned from a warm, dynamic world to the frozen desert today.

²⁰) In the region between proto-Mars and proto-Jupiter, billions of small, rocky and icy chunks called planetesimals begin to condense. As Jupiter fully forms, its immense gravitational pull begins to disrupt the region, the 'Jupiter Interruption', instead of gently fusing into a single new planet, the planetesimals collide violently and shatter.

²¹) According to planetary migration models like the 'Grand Tack', Jupiter moves inward toward the Sun and then back out. This cosmic sweeping flings dry, rocky bodies from the inner Solar System outward, and icy, carbon-rich bodies from the outer Solar System inward, mixing them together into the belt today.

²²) A major orbital shift among the giant planets (the 'Late Heavy Bombardment' or 'Nice Model' period) destabilizes the belt. Gravitational resonances eject over 99% of its original mass, hurling rocks into the inner Solar System and heavily cratering Earth and the Moon.

²³) Jupiter, the oldest planet, forms a solid core of silicates and ice beyond the solar system's 'snow line' and begins rapidly vacuuming up surrounding hydrogen and helium gas.

²⁴) Jupiter's interaction with the solar gas disk drags the planet inward toward the Sun (to where Mars sits today). Saturn's gravitational pull later drags Jupiter back out to its current position.

²⁵) Jupiter develops its own disk of debris, which 'condenses' into its four largest Galilean moons: Io, Europa, Ganymede and Callisto.

²⁶) Jupiter and the other gas giants shift orbits, this destabilizes the Solar System, triggering a period of chaotic asteroid impacts known as the 'Late Heavy Bombardment'.

²⁷) Recent scientific consensus suggests Saturn's ring system may have formed dynamically during this window from a destroyed icy moon.

²⁸) Uranus forms from the solar nebula as a massive 'ice giant' composed primarily of water, ammonia and methane.

²⁹) An Earth-sized planetesimal collides with Uranus, which massive impact permanently knocks the planet onto its side, giving it an extreme 97.8° axial tilt and a retrograde rotation.

³⁰) Jupiter and Saturn shift orbits, pushing Uranus and Neptune further out into the frozen edges of the Solar System.

³¹) Gravitational interactions disrupt some inner moons, the debris shatters to form Uranus' narrow, dark inner ring system.

³²) Icy planetesimals fail to coalesce into a large planet due to low material density at the outer edge.

³³) Neptune moves outward, scattering up to 99% of the primordial belt's original mass into deep space and creating the 'scattered disc'.

³⁴) The Hills cloud begins forming alongside the solar system. Giant planets gravitationally scatter icy planetesimals outward into distant orbits.

³⁵) Galactic tides and passing birth-cluster stars circularize these chaotic orbits, which creates a dense, stable inner disk.

³⁶) It constantly repopulates the easily disrupted outer Oort cloud with trillions of icy bodies.

³⁷) The young Sun is ejected from its native birth cluster and begins to 'navigate' the Milky Way galaxy as an isolated, independent planetary system.

³⁸) As the Solar System leaves its cluster, the massive gravitational influence of neighboring stars fades. The 'galactic tide', forces exerted by the Milky Way's disk, and passing stars stabilize the thrown-out debris, which shapes the scattered icy bodies into the spherical outer Oort Cloud.

³⁹) It captures comet-like objects originally belonging to other stellar systems that pass too close to the Sun.